

Response to Reviewers — IEEE OMLET 2026, Paper ID 528

Title: Leakage-Aware Intersectional Fairness Mitigation for Distributed Rating Classification in Recommender Systems

We thank both reviewers for the careful reading and the constructive comments. The decision was *accept with minor revision*. Below we address every comment and point to the exact location of each change in the revised manuscript. All changes are textual (framing, scope, and limitations); no results were altered.

Reviewer 1

R1.1 — “The paper lacks technical novelty as threshold calibration is standard.”

We agree that threshold calibration is a standard post-processing tool and we do not claim it as a new technique. We have made the actual contribution explicit and moved it to the front of the paper.

- **Added** a new “*Scope and novelty*” paragraph immediately after the Contributions list (§I). It states that the contribution is the *combination* prior work leaves unaddressed — a leakage-aware fairness audit on a **distributed** rating pipeline, an **intersectional** calibration benchmarked head-to-head against single-attribute calibration **with paired significance tests**, and replication across an independent dataset and a second model class.
- This complements the existing Related Work text (§II, “We do not claim intersectional fairness as novel; rather, we *adapt* threshold calibration to the joint attribute cell ...”) and the positioning Table I, which shows no prior entry combining all four properties.

R1.2 — “The evaluation relies on small public datasets with demographic information.”

- **Revised** the *External validity* paragraph (§VIII) to note that MovieLens 1M and 100K are the de-facto benchmarks for demographic fairness in recommendation (Ekstrand et al.), and that public rating datasets carrying self-reported gender and age are scarce because such attributes are seldom collected or released for privacy reasons. This constrains cross-domain evaluation for the subfield as a whole, and we now say so explicitly rather than treating it as a one-line caveat.
- The study already validates every headline result across eight random splits, a second independent dataset (§VII), and a second model class (gradient-boosted trees, §VII-A), which is the strongest form of external check available given the data landscape.

R1.3 — “The intersectional calibration method fails to scale with multiple attributes.”

We agree, and the manuscript already analyzes this explicitly.

- §VIII, *Scalability* paragraph: with d attributes of k levels the number of calibrated cells grows as k^d ; beyond a few attributes per-cell samples shrink until quantile estimates become unstable. We explicitly state the results “should be read as evidence for *joint* over single-attribute post-processing at low attribute counts, **not as a claim of scalability to high-dimensional intersections**,” and point to subgroup-fairness methods (Kearns et al.) as the direction for many attributes.

- §IX (Conclusion / Future Work) lists regularized or hierarchical intersectional thresholds for many fine-grained strata as the follow-up.

No change required; we note the scope was already bounded in the submitted version.

R1.4 — “Target leakage correction yields only a negligible difference in performance.”

This is correct on aggregate accuracy ($0.723 \rightarrow 0.719$), and that is precisely the point — the correction is not aimed at predictive gain. We have clarified this.

- **Revised** §I (“First, ...”): the correction is necessary “not [for] predictive performance but [for] the integrity of the per-group fairness metrics computed afterward, since leaked targets contaminate the same selection rates the audit inspects.”
- **Revised** §V, *Leakage correction*: leaked features embed the target in the input, so a fairness audit run on the leaky model reports per-group selection rates and disparity measures that are partly an evaluation artifact rather than a property of the model. Leakage here distorts the **fairness** reading, not the headline score.

R1.5 — “The study audits binary rating prediction instead of recommendation ranking.”

This is a deliberate scoping choice, now stated up front.

- **Added** to the new *Scope and novelty* paragraph (§I): we audit the rating-classification stage rather than top-k ranking because score bias at this stage propagates into any ranking built on it, so the classifier is where a disparity must first be measured and corrected.
- Existing text already frames this (§II, “we audit the upstream rating-classification stage, where disparities originate but are rarely measured”) and §IX lists “extending the audit from rating classification to top-k recommendation” as future work.

Reviewer 2

R2.1 — “The leakage-aware evaluation is carefully designed and strengthens the credibility of the fairness analysis.”

We thank the reviewer. The clarifications made for R1.4 further sharpen why the leakage-aware design matters for the fairness reading specifically.

R2.2 — “The findings are well validated across multiple random splits, a second dataset and another classifier.”

We thank the reviewer; no change.

R2.3 — “Generalization remains limited because both evaluation datasets are from the MovieLens domain.”

We agree and have strengthened the acknowledgement.

- **Revised** §VIII *External validity* (same edit as R1.2): generalization to non-movie domains remains open; we now frame this as a subfield-wide constraint driven by the scarcity of demographically labeled public rating data, and we do not generalize beyond the audited setting without re-auditing.

Summary of changes

Location	Change
§I, after Contributions	New “ <i>Scope and novelty</i> ” paragraph (R1.1, R1.5)
§I, “First . . .” sentence	Reframed leakage correction around fairness-metric integrity (R1.4)
§V, <i>Leakage correction</i>	Added rationale: leakage distorts the fairness reading, not accuracy (R1.4)
§VIII, <i>External validity</i>	Benchmark justification + subfield-wide data-scarcity framing (R1.2, R2.3)
Back matter	<i>AI Use Disclosure</i> section added; <i>Reproducibility</i> wording tightened (camera-ready)

No experimental results, tables, or figures were changed. The manuscript remains 6 pages.